

HeLLMholtz & OpenCode Manual

Helmholtz LLM Suite Guide for Colleagues

Helmholtz Research Team

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Chapter 1

Quick Start

1.1 HeLLMholtz CLI

```
# Install
pip install "hellmholtz[eval,proxy]"

# Configure once
hellm setup setup --api-key "YOUR_BLABLADOR_TOKEN" \
                  --base-url "https://api.blablador.fz-juelich.de/v1"

# Chat with a model
hellm chat --model blablador:alias-code "Explain quantum computing"

# List available models
hellm models

# Run benchmarks
hellm bench --models blablador:alias-code,blablador:alias-fast \
            --prompts-category reasoning
```

1.2 OpenCode AI Assistant

```
# Install OpenCode
curl -fsSL https://opencode.ai/install | bash

# Run
opencode

# Single query
opencode "Write a Python function to calculate fibonacci numbers"
```

Chapter 2

What is HeLLMholtz?

HeLLMholtz is a unified LLM access, benchmarking, and evaluation suite built for Helmholtz researchers.

2.1 Key Features

- **Unified API:** Access multiple LLM providers through a single interface
- **Benchmarking:** Run standardized benchmarks across models
- **Evaluation:** Automated evaluation with LLM-as-a-Judge scoring
- **Throughput Testing:** Measure token generation speed
- **Model Monitoring:** Track availability and performance

Chapter 3

What is OpenCode?

OpenCode is a private, open-source CLI AI coding assistant.

3.1 Why OpenCode?

Feature	Claude Code	OpenCode
Privacy	Sends to OpenAI	Runs on your infrastructure
Cost	Pay per token	Free (self-hosted)
Control	Limited	Full control
Data	External	Internal only

3.2 Blablador Models

Model	Context	Output	Use Case
alias-code	98K	32K	Primary coding assistant
alias-fast	98K	32K	Quick responses
alias-large	98K	32K	Large context handling
alias-huge	98K	32K	Maximum capability

Chapter 4

Installation & Setup

4.1 System Dependencies

```
# Install pipx (for CLI tools)
pip install pipx
pipx install poetry
```

4.2 Get Your Blablador Token

1. Visit: <https://codebase.helmholtz.cloud>
2. Sign in with Helmholtz ID
3. **User Settings** → **Access** → **Personal Access Tokens**
4. Click **New Token**
5. Name: Blablador API Token, Scope: read_user
6. Copy the token (shown only once!)

4.3 Configure HeLLMholtz

```
# Install the package
pip install "hellmholtz[eval,proxy]"

# Configure (persistent across projects)
hellm setup setup --api-key "YOUR_TOKEN" \
                 --base-url "https://api.blablador.fz-juelich.de/v1" \
                 --show # View current config
```

Configuration saved to: ~/.config/hellmholtz/.env

4.4 OpenCode Configuration

OpenCode is pre-configured at ~/.config/opencode/opencode.json with: - Primary model: blablador/alias-code - Small model: blablador/alias-fast - MCP servers:

Obsidian, Markitdown, Docling, Europe PMC

Chapter 5

HeLLMholtz Usage

5.1 Chat Interface

Simple chat

```
hellm chat --model blablador:alias-code "Explain quantum computing"
```

With system prompt

```
hellm chat --model openai:gpt-4o \  
    --system "You are a Python expert" \  
    "How do I use decorators?"
```

Interactive mode

```
hellm chat --model blablador:alias-code --interactive
```

5.2 Model Management

List available models

```
hellm models
```

Get model details

```
hellm models --details
```

Monitor model health

```
hellm monitor
```

Test accessibility

```
hellm monitor --test-accessibility
```

5.3 Benchmarking

Using prompt categories

```
hellm bench --models blablador:alias-code,openai:gpt-4o \  
    "Explain quantum computing"
```

```
    --prompts-category reasoning

# All categories
hellm bench --models blablador:alias-code --all-prompts

# Custom prompts
hellm bench --models blablador:alias-code \
    --prompts-file custom_prompts.json

# Advanced benchmarking
hellm bench --models blablador:alias-code,blablador:alias-fast,openai:gpt-4o \
    --prompts-file custom_prompts.json \
    --prompts-category reasoning \
    --temperatures 0.1,0.7,1.0 \
    --replications 3 \
    --evaluate-with openai:gpt-4o \
    --results-dir results/
```

5.4 Throughput Testing

```
hellm bench-throughput --model blablador:alias-code \
    --requests 100 \
    --concurrency 10
```

5.5 Reporting & Analysis

```
# Markdown report
hellm report --input-file results/benchmark.json \
    --output-file report.md

# HTML report with charts
hellm analyze results/benchmark_latest.json \
    --html-report analysis_report.html
```

Chapter 6

OpenCode Usage

6.1 Running OpenCode

Interactive session

```
opencode
```

Single query

```
opencode "Write a Python function to calculate fibonacci numbers"
```

6.2 Using Different Models

In OpenCode session, switch models

```
/model blablador:alias-huge
```

6.3 Context Management

OpenCode uses Dynamic Context Pruning (DCP) automatically:

```
"compaction": {  
  "auto": true,  
  "prune": true,  
  "reserved": 10000  
}
```

6.4 Obsidian Integration

Connects to your coding-brain vault for: - Knowledge management - Context preservation - Wikilinks between concepts

Chapter 7

Agent System

7.1 Available Agents

Agent	Purpose
paper-assistant	Primary coordinator for paper tasks
paper-architect	Structure and organization
academic-writer	Writing quality and drafting
latex-wizard	LaTeX formatting
citation-manager	Bibliography management
figure-creator	Creating plots and diagrams
literature-reviewer	Literature search

7.2 Using Agents

In OpenCode session:

```
@paper-architect create an outline for a paper on quantum computing
@academic-writer draft the introduction section
@latex-wizard format this section in LaTeX
@citation-manager check my .bib file for duplicates
@figure-creator create a plot of the experimental results
@literature-reviewer find recent papers on quantum computing
```

Chapter 8

Best Practices

8.1 Model Selection

Task	Recommended Model
Programming	alias-code
Quick questions	alias-fast
Long documents	alias-large
Complex reasoning	alias-huge

8.2 Context Management

- Keep conversations under 100K tokens
- Use `--interactive` mode for long sessions
- Clear context periodically with `/clear`

8.3 Benchmarking Tips

- Test multiple temperatures (0.1, 0.7, 1.0)
- Run at least 3 replications for reliability
- Use `--evaluate-with` for LLM-as-a-Judge scoring
- Compare against baseline models

Chapter 9

Troubleshooting

9.1 Common Issues

Problem	Solution
API key not set	Run <code>hellm setup setup --api-key KEY</code>
Model not found	Check <code>hellm models</code> for available models
Token limit exceeded	Use shorter prompts or split messages
Connection timeout	Check internet and <code>curl</code> https://api.blablador.fz-juelich.de/v1/models

9.2 Verification Commands

```
# Test Blablador connection
```

```
hellm models
```

```
# Test OpenCode connection
```

```
opencode "Say hello"
```

```
# Check configuration
```

```
hellm setup setup --show
```

Chapter 10

Quick Reference

10.1 HeLLMholtz CLI Commands

Task	Command
Chat	<code>hellm chat --model blablador:alias-code "prompt"</code>
List models	<code>hellm models</code>
Monitor	<code>hellm monitor --test-accessibility</code>
Benchmark	<code>hellm bench --models ... --prompts-category reasoning</code>
Report	<code>hellm report --input-file results.json --output-file report.md</code>
Analyze	<code>hellm analyze results.json --html-report analysis.html</code>

10.2 OpenCode Commands

Task	Command
Start	<code>opencode</code>
Single query	<code>opencode "prompt"</code>
Switch model	<code>/model blablador:alias-fast</code>
Use agent	<code>@paper-architect create outline</code>
Clear context	<code>/clear</code>

Chapter 11

Resources

11.1 Documentation

- **HeLLMholtz:** `/home/jhe24/AID-PAIS/helmholtz_llm_suite/docs/`
- **Blablador:** <https://blablador.fz-juelich.de>
- **OpenCode:** <https://opencode.ai>

11.2 Configuration Files

- **HeLLMholtz:** `~/.config/hellmholtz/.env`
- **OpenCode:** `~/.config/opencode/opencode.json`
- **Agent files:** `~/.config/opencode/agents/`
- **Obsidian vault:** `/home/jhe24/Documents/coding-brain/`

11.3 Important Paths

- **Project root:** `/home/jhe24/AID-PAIS/helmholtz_llm_suite`
 - **Source code:** `/home/jhe24/AID-PAIS/helmholtz_llm_suite/src/hellmholtz/`
 - **Model config:** `/home/jhe24/AID-PAIS/helmholtz_llm_suite/src/hellmholtz/providers/blablador_c`
-

11.4 Summary

This guide covers:

- **HeLLMholtz:** Unified LLM access, benchmarking, and evaluation
- **OpenCode:** Private AI coding assistant with Blablador
- **Blablador:** Helmholtz's LLM infrastructure
- **Agent System:** Specialized AI agents for paper writing
- **Best Practices:** How to use everything effectively

Happy coding with Blablador!